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Conversion efficiencies of electron beam energy to vacuum ultraviolet light for Ne, Ar, Kr, and Xe excited with continuous electron beams

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Continuous 8–15 keV electron beams were used to excite Ne, Ar, Kr, and Xe at pressures ranging from 75 to 1400 hPa. Conversion efficiencies of the energy deposited by the electron beam in the gas to vacuum ultraviolet light emitted from excimers are obtained. At pressures above 500 hPa, the efficiencies show constant values of 0.31 ± 0.06 for Ne, 0.33 ± 0.04 for Ar, 0.42 ± 0.05 for Kr, and 0.42 ± 0.05 for Xe. Apparent energy conversion efficiencies for practical devices utilizing this excitation technique are also reported. © *2008 American Institute of Physics*.
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I. INTRODUCTION

Excitation of rare gases at pressures of ~100 hPa and above results in a strong broadband emission in the vacuum ultraviolet (VUV) spectral region.[1] This emission originates from excited diatomic molecules (excimers) in the lowest excited states. These molecular states are directly populated from the two lowest excited atomic states which have long effective radiative decay times. The broadband emission is weakly affected by radiation trapping since it originates from the transitions to the repulsive molecular ground state. Because of these properties of rare gas excimers, a high conversion efficiency of the excitation energy deposited in the gas to the VUV light can be expected. Moreover, rare gases are therefore an attractive amplifying medium for VUV lasers.

VUV emission from rare gases has been investigated in a large number of studies. Gas excitation was performed with various types of discharges[2–10] as well as with electron beams.[11–16] Many papers report measured values or theoretical calculations of the conversion efficiency of the energy deposited in the gas to the VUV emission. The highest efficiency of 60% has been reported for dielectric barrier discharges in Xe.[8,9] Efficiencies of ~50% were measured for corona discharges,[4,5] up to 40% for microwave discharges,[10] and up to 20% for microhollow cathode discharges.[6]

For electron beam excitation of rare gases, theoretical calculations of the efficiencies show values of about 50%.[13,14,17] Experimental measurements were typically performed with pulsed, high energy (at least several hundreds keV), and high current (~1 kA) electron beams, usually applied in studies of light amplification in rare gases. Values close to the maximum theoretical efficiencies have been reported for low excitation power.[12–14] The published efficiency results, however, show large uncertainties due to calculations of the geometrical factor of the observed VUV emission as well as measurements of the gas heating caused by the deposition of the electron beam energy. Information on the gas heating was required to calculate the total power deposited by the beam in the target gas. Another problem that is difficult to overcome in such large scale and high power installations is the high rate of impurity production in the target cell.

Here we present results of the energy conversion efficiencies for relatively low energy (8–15 keV), low current (~5 μA) continuous electron beam excitation of Ne, Ar, Kr, and Xe in the pressure range of 100–1400 hPa. This low energy electron beam excitation technique has already been successfully applied in light sources[15,16,18,19] and gas kinetic studies.[20–22] Also, intrinsic properties of such excitation allow another approach to the measurements of the conversion efficiencies, producing data with lower uncertainties than those in a high energy beam apparatus. In this study, because of the short penetration depth of the low energy electrons, it was possible to assume a point light source geometry for most of the target gas pressures. The low power of the electron beam and a fast circulation of the target gas through a rare gas purifier resulted in a very low level of impurities. Moreover, the small size of the light emitting volume allowed continuous wavelength-resolved monitoring of the light emission from the whole excitation region. Application of the windowless spectroscopic technique[16] extended the spectroscopic observations to the extreme ultraviolet (XUV) wavelength range. The latter made it possible to report efficiencies for Ne, which, to our best knowledge, has not been done before.

Calculations of the energies deposited by the electron beam in the gas and in the entrance foil as well as the energy lost due to backscattering of electrons have been performed with the GEANT4 package.[23,24] GEANT4 is a toolkit for simulation of the passage of particles through matter developed at CERN (The European Organization for Nuclear Research). Due to the recent addition of the low energy electromagnetic physics package, the GEANT4 code is now applicable for particle energies relevant for this study.

Apparent energy conversion efficiencies, which are defined below and contain only directly measurable quantities, are discussed in Sec. III A. Calculations of the physical efficiencies of the energy conversion are given in Sec. III B.

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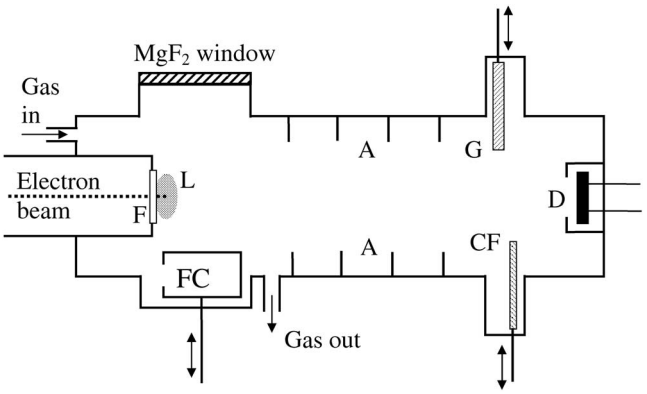


FIG. 1. Schematic drawing of the experimental setup. F is the ceramic foil, L is the excited volume of the gas, FC is the Faraday cup, A is the aperture assembly, CF is the carbon foil filter, D is the detector, and G is the glass filter. Emission spectra were recorded with a VUV monochromator (not shown) through the MgF_2 window. This window was removed for the XUV spectroscopic observations.

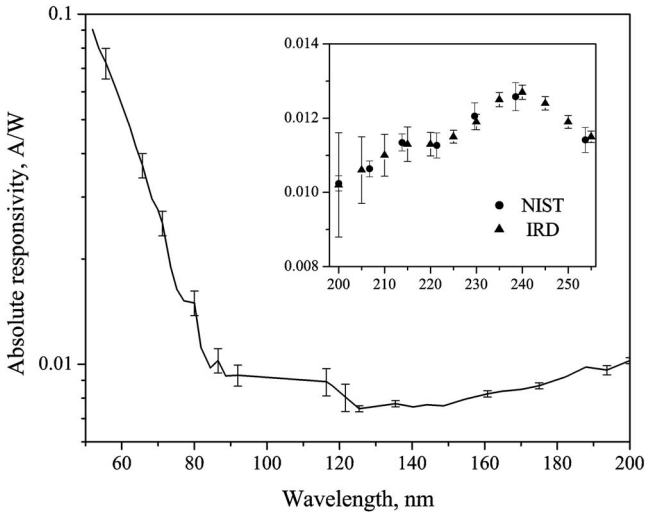


FIG. 2. Main graph: The calibration curve supplied by NIST. Insert: The calibration curves from NIST and IRD in the overlap range.

II. EXPERIMENTAL SETUP

Excitation of rare gases was performed with continuous (~5 μA, 8–15 keV) electron beams which were produced with an electron gun. A description of a similar electron gun and its electrical circuits is given in our previous report.[21] The electron beams were sent into a gas cell through a thin ceramic foil. The foil had two layers: 50 nm SiO_2 on the side facing the electron gun and 250 nm Si_3N_4 on the target gas side. These two layers were manufactured on a silicon substrate by chemical vapor deposition. The substrate material was removed by chemical etching to form square foils of 0.5×0.5, 0.7×0.7, or 2×2 mm^2. The silicon substrate around each foil had been cleaved to 5×5 mm^2 frames. Note that the whole surface of the frame facing the gas was covered by silicon nitride.

Target gas pressures ranged from 75 to 1400 hPa. The gas cell (see Fig. 1) was a stainless steel cylinder with an inner diameter of 37 mm. The target gas was continuously circulated through the cell and a rare gas purifier (MonoTorr, Saes getters) with a metal bellows compressor. Rare gases with purity of at least 99.9998% prior to purification were used.

Electron beam currents were measured in vacuum with a Faraday cup. The cup was installed 0.5 mm away from the foil on a mechanical feedthrough, which allowed retraction during light emission measurements. The cup current was measured with an electrometer (Keithley, model 614). When no foil was installed between the electron gun and the evacuated gas cell, the Faraday cup current was equal (difference was less than 1%) to the cathode current of the gun, i.e., all electrons produced by the gun were detected by the Faraday cup. After mounting a foil, a reduction of 12%–15% in the cup current was observed, which is in very good agreement with our simulations of the electron beam passage through the foil performed with GEANT4. Since the foil produces a significant angular straggling of the low energy electron beams, it was essential to confirm that all electrons scattered in the foil were detected by the cup. Therefore, another temporary Faraday cup was installed. It was possible to steer this cup along the electron beam direction. Measurements of the current when the foil and its silicon frame were completely inside the cup have shown that the differences between the current readings given by both cups were less than 2%.

A semiconductor detector was used for absolute intensity measurements. The detector (SXUV-100, IRD Inc.) was calibrated in the range from 200 to 1100 nm by the manufacturer and then sent to NIST (National Institute of Standards and Technology) for calibration in the range of 54–254 nm (see Fig. 2). In the overlap region (200–250 nm), both calibrations show an excellent match. The detector was installed inside the gas cell facing the electron beam at the distance of 152 mm to the entrance foil. Six apertures were installed inside the gas cell to prevent detection of light reflected by the walls. An additional aperture (8 mm in diameter) was installed directly in front of the detector to screen the uncalibrated rim region of the detector.

The photocurrent was measured with an electrometer (Keithley, model 614). To cross-check the detector calibration and the photocurrent readout, the electron gun and the foil were temporally replaced by a calibrated halogen lamp. The measured photocurrent and the value calculated using the intensity calibration of the lamp and the detector response curve showed a difference of less than 5%.

A retractable glass filter was used to separate contributions to the photocurrent originating from the visible-infrared and the VUV regions. According to the measurements with a halogen lamp, the filter was completely blocking light with wavelengths $\lambda<280$ nm and had a transmission of 91% for $\lambda>380$ nm.

Light emission from rare gases was also analyzed spectrally. A MgF_2 window was installed in the side wall of the gas cell. Light output in the wavelength range of 116–400 nm was resolved with a McPherson monochromator (model 218) and detected with a photomultiplier (PM) tube with an S20 cathode. The PM tube was operated in photon-counting mode. A relative intensity calibration of the monochromator and the PM tube had been performed using a deuterium lamp calibrated by PTB (Physikalisch-Technische

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Bundesanstalt). To record the XUV light output from neon, a windowless spectroscopic technique[16] was used, covering the wavelength range from 60 to 200 nm. In this case, both the gas cell and the monochromator were filled with neon. The XUV light resolved by a monochromator was converted to visible light by using a sodium salicylate wavelength shifter in front of the PM tube.

III. RESULTS AND DISCUSSION

Energy conversion efficiency is defined here as the ratio of the power emitted by the target gas into 4π solid angle in the VUV (or XUV) wavelength range and the power deposited in the gas by the electron beam. However, there are several factors which disallow (or strongly hinder) direct measurements of the quantities involved in this definition. These factors are the backscattering of electrons from the gas, electron energy loss in the foil, reflection of the VUV light on the silicon nitride surface, bulging of the foil due to the pressure differential, and variation in the geometry of the light emitting volume with the gas pressure. Therefore, we define a second quantity, an "apparent" energy conversion efficiency, which is used here as an intermediate step towards the calculation of the energy conversion efficiency (called "physical" efficiency below). The apparent energy conversion efficiency η_a is defined as

$$\eta_a = W_d \frac{4\pi d^2}{A_d} \frac{1}{EI_F}, \tag{1}$$

where W_d is the total energy of VUV photons which reach the detector per second, d is the distance between the foil and the detector (assumed to be large enough for point-source approximation), A_d is the effective area of the detector, E is the particle energy of electrons (in eV) provided by the electron gun, and I_F is the current of the electron beam after passing through the foil. Note that all quantities involved in Eq. (1) are directly measurable. Experimental results and apparent efficiencies for Ne, Ar, Kr, and Xe are discussed in Sec. III A and calculations of corresponding physical efficiencies are presented in Sec. III B.

A. Apparent efficiencies

VUV to ultraviolet (UV) spectra of Ar, Kr, and Xe at 1200 hPa gas pressure are shown in Fig. 3. The main emission is the second continuum of the corresponding excimer. The spectra are corrected for the spectral response of the detection system. Note that the intensity of impurity lines is extremely low. Figure 4 shows an emission spectrum from Ne at 1200 hPa. The relative intensities are corrected for the grating response provided by the manufacturer. It is also taken into account that the quantum conversion efficiency of the sodium salycilate wavelength shifter is nearly constant in this wavelength region.[25] The peak intensity of the hydrogen Lyman-α line is rather strong, however, its wavelength integrated contribution to the XUV/VUV emission intensity is less than 5%. Therefore, we have ignored the presence of the Lyman-α emission in calculations of the apparent efficiencies. However, an additional uncertainty associated with this impurity emission has been taken into account for the Ne efficiency data (discussed below). The Lyman-α emission appears due to energy and charge transfer to the hydrogen impurity which cannot be effectively removed by the titanium rare gas purifier.[22] Other impurity lines in the VUV and UV regions were negligible.

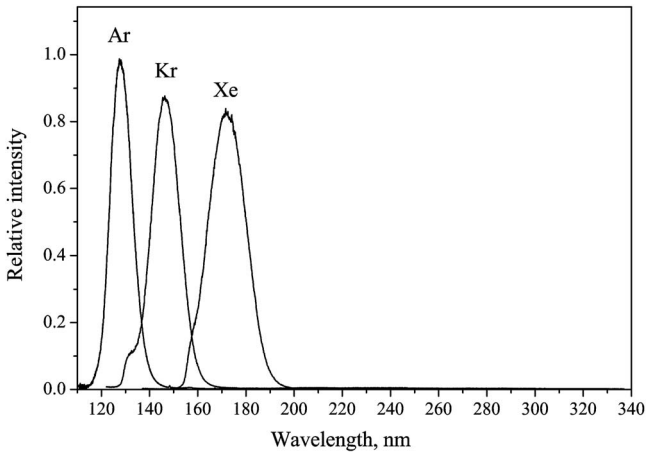


FIG. 3. Emission spectra of Ar, Kr, and Xe at 1200 hPa gas pressure. Note the absence of impurity lines.

In helium the second continuum emission has generally low intensity. In our conditions, the integrated intensity of the He second continuum was comparable to the intensity of the Lyman-α line. Helium data are therefore not presented in this paper.

As described in Sec. II, a calibrated semiconductor detector was used to measure absolute intensities of the light emission. The photocurrent due to the VUV emission, which is directly proportional to the VUV intensity, was calculated as

$$J = J_{\mathrm{NF}} - \frac{J_{\mathrm{F}}}{T_{\mathrm{F}}}, \tag{2}$$

where J_{NF} and J_{F} are the photocurrents without and with the glass filter, respectively, and T_{F} is the transmission of the

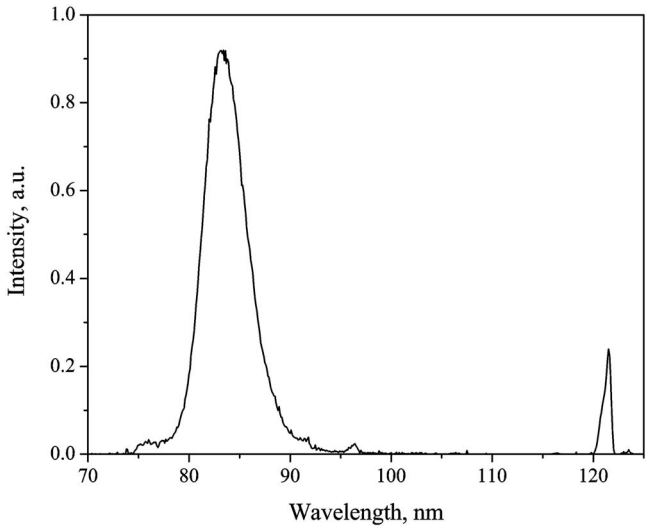


FIG. 4. Emission spectrum of Ne at 1200 hPa gas pressure. The Lyman-α line (~121.6 nm) from a hydrogen impurity contributes about 5% to the XUV/VUV intensity.

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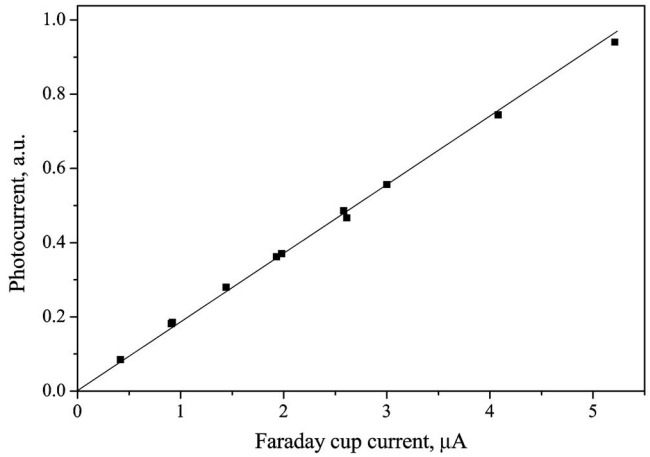


FIG. 5. Photocurrent of the semiconductor detector due to the VUV emission as a function of electron beam current for 1000 hPa of Ar (dots). A linear fit is also shown.

filter in the visible range. However, this formula gives accurate values of the photocurrent due to the VUV emission only if two conditions are fulfilled. The integrated intensity of impurity lines in the near UV region where the glass filter has a transmission of less than T_F should be negligible. Contribution of soft x rays generated in the foil or in the gas to the photocurrent has to be negligible too. Figure 3 shows that the first condition is satisfied. According to the x-ray absorption data,[26] there should be zero contribution to the photocurrent from x rays when Xe is used as the target gas. However, for lighter gases, especially Ne at low pressures, the most energetic (at least a few keV) x-ray photons could reach the detector. In order to check their potential contribution to the photocurrent, a thin carbon foil filter was used. This filter was transparent in the x-ray region and completely (to less than 0.05% of the original intensity) blocking the VUV and the visible light. Measurements of the photocurrent with the carbon filter have shown that in Xe, the x-ray contribution to the photocurrent was completely absent. In Ne the photocurrent due to the x rays was negligible as well: its amplitude was less than 2% of the photocurrent due to the VUV emission.

The variation in the photocurrent due to the VUV emission with the electron beam current is shown for Ar at 1000 hPa in Fig. 5. The linear dependence of the photocurrent shows that for a fixed particle energy of the electron beam, the conversion efficiency is independent of the excitation power in the investigated range of 5–50 mW.

The total energy of the VUV photons which reach the detector per second was calculated from the photocurrent values by using the wavelength-resolved quantum efficiency calibration of the semiconductor detector and the spectral shape of the excimer emission. The resulting pressure dependencies of the apparent efficiency for Ne, Ar, Kr, and Xe calculated according to Eq. (1) are presented in Fig. 6 for the particle energy of the beam of 12 keV. The figure shows that at pressures above 500 hPa, the efficiencies are nearly constant and have values of 0.30 ± 0.05 for Ne, 0.31 ± 0.03 for Ar, 0.35 ± 0.02 for Kr, and 0.35 ± 0.02 for Xe. The error bars are the maximum systematic uncertainties which appear due to the emission intensity measurements (10% for Ne, 5% for Ar, and 3% for Kr and Xe) and the beam current measurements (3%). An extra 5% uncertainty is added for Ne due to the hydrogen impurity. This value appears due to the following reasons: Taking into account the corresponding calibrations, the maximum overestimation of the conversion efficiency is equal to the relative contribution of the Lyman-α emission to the recorded XUV intensity (5%). The maximum underestimation is also about 5%: The minimum value for the apparent conversion efficiency of the electron beam energy to the Lyman-α emission which has been found for $Ne-H_2$ mixtures[27] is about two times lower than the apparent conversion efficiency derived for Ne in this study. At

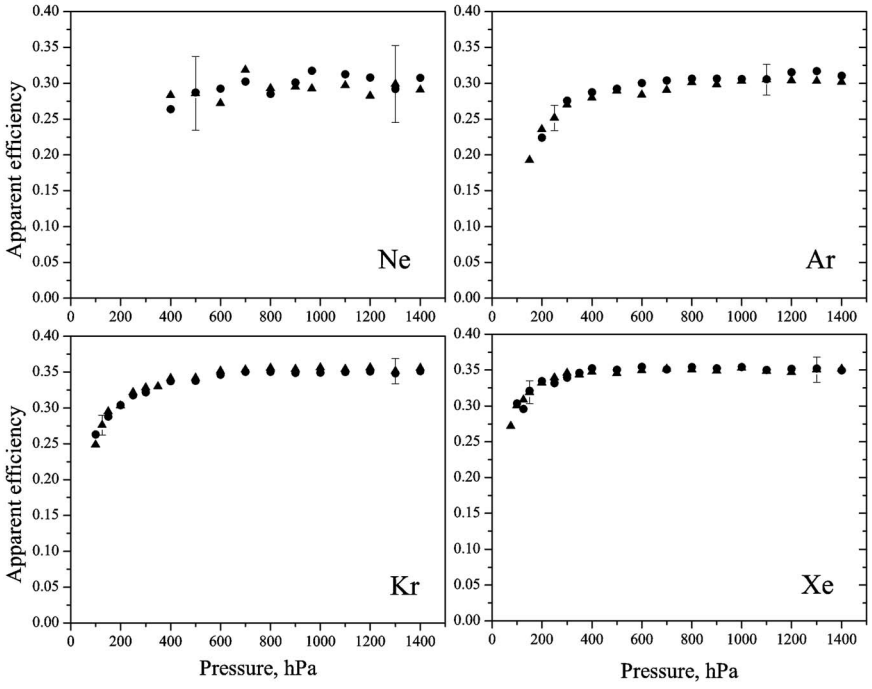


FIG. 6. Apparent conversion efficiencies of electron beam energy to VUV light. Apparent efficiency is calculated by ignoring energy loss in the entrance foil and assuming a pointlike light source. Data are shown as a function of gas pressure for Ne, Ar, Kr, and Xe excited by 12 keV electron beams using a 50 nm SiO_2+250 nm Si_3N_4 entrance foil.

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these conditions, only half of the intensity lost by the excimer emission will reappear in the Lyman-α emission. Note that we assumed a 30%—50% transmission of the MgF_2 window used in the cited $Ne-H_2$ study. A discussion of energy transfer channels in $Ne-H_2$ mixtures is given in our previous report.[22]

The statistical uncertainty in the data presented in Fig. 6 is about 5% (~10% for Ne data). It originates from fluctuations in the electron beam current, beam steering, and focusing. The statistical uncertainty is largest for Ne since it is only for this gas that the visible light intensity is strong in comparison with the VUV intensity.

An increase in the particle beam energy from 12 to 15 keV results in a ~10% increase in apparent efficiency values. A reduction in the energy from 12 to 8 keV leads to an ~25% decrease in apparent efficiencies. These variations, as will be discussed in the next section, are mainly due to loss of the electron beam energy in the foil.

Additional measurements have been performed to show the consistency of the obtained efficiency values. It was shown that the apparent efficiency changes by less than 5% when the distance between the foil and the detector varies within 152 and 277 mm. Also, efficiency measurements have been performed with a 3° offset between the electron beam direction and the line connecting the centers of the foil and the detector. The resulting efficiency values have been the same as with a 0° offset.

B. Physical efficiency

As mentioned in the beginning of Sec. II, a number of factors have to be taken into account in order to convert the apparent efficiencies obtained above into physical efficiencies, which accurately account for the energy emitted by the excited gas and the energy deposited in the gas by the beam. All these factors require numerical simulations which trace electrons in the foil and in the gas. In this study, we have used two simulation programs: GEANT4 package[23,24] and CASINO V2.42.[28] Previously we have shown[29] that CASINO can rather accurately simulate spatial distributions of emission from dense gases. Therefore this program has been used here when the geometry of the light source was concerned. CASINO can also provide relative energies (per one electron of the beam) dissipated in the gas, in the foil, and lost due to backscattering. Unfortunately, by using this program, it is not possible to take into account that some of the primary electrons are backscattered from the foil and do not penetrate into the gas. These electrons would not be recorded with the Faraday cup and should be disregarded from the analysis. Therefore, the more flexible GEANT4 code has been used to calculate the average energy deposited in the target gas per primary electron. By using the low energy electromagnetic physics package, GEANT4 allows individual tracing of all particles, including secondary electrons which are created with at least 100 eV kinetic energy. Ionization events which result in production of a secondary electron with an energy less than 100 eV are treated as a "continuous" energy-loss process without individual tracing of secondaries. Note that GEANT4 traces each particle until it has zero energy.

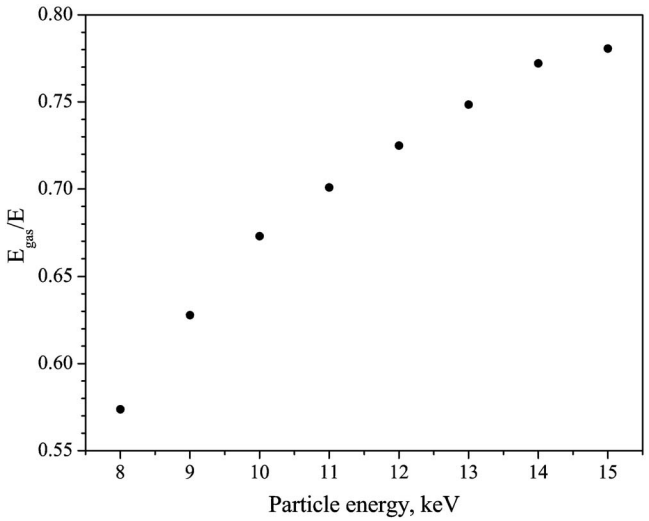


FIG. 7. Ratio of the energy E_{gas} deposited in the gas (per primary electron) and the particle energy E of the beam before the interaction with the foil, calculated with GEANT4 for Ar.

The apparent energy conversion efficiencies contain the particle energy of electrons before the interaction with the foil. Therefore, in order to calculate the physical efficiency, the values of the apparent efficiencies should be divided by the ratio of the average energy deposited in the gas per primary electron and the particle energy of the electron beam before interacting with the foil. GEANT4 simulations show that these ratios are independent of the gas pressure but vary significantly with the target gas due to backscattering from the gas into the foil. For an electron beam with 12 keV particle energy, the ratios are 0.68 for Xe, 0.67 for Kr, 0.73 for Ar, and 0.76 for Ne. The uncertainty in these values is about 10% (discussed below). The ratios strongly depend on the particle energy. An example of the energy dependence is given for Ar in Fig. 7. Since the physical efficiency is expected to be nearly constant in this energy range (see the discussion[30] of Ar excitation versus particle energy), the values of the apparent efficiency corrected for the energy deposited in the gas should show no dependence on the particle energy of the beam if GEANT4 performs an accurate tracing of electrons. As demonstrated in Fig. 8 for Ar at 1000 hPa, the corrected apparent efficiency shows indeed very small variations with the particle energy of the beam. Note that for Ar at this pressure, all other corrections which have to be applied to the apparent efficiency in order to produce the physical efficiency (discussed below) are practically independent of the electron beam energy in this energy range.

The second factor which has to be considered is the light reflection on the silicon nitride surface (see Sec. II for the geometry details). We have measured the reflection coefficient at 45° angle of incidence on a flat silicon sample covered with the silicon nitride layer. This sample and all the foils used in this study were taken from the same silicon wafer. The results are shown in Fig. 9 together with the values calculated with the Fresnel formula using the refraction index and extinction coefficients reported in the literature.[31] Since there is good agreement between the calculations and the measured values, we have used the Fresnel formula to

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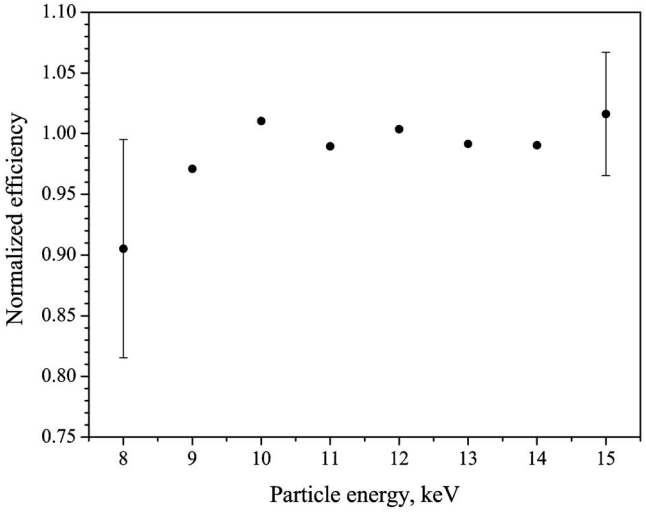


FIG. 8. Normalized apparent efficiencies for Ar at 1000 hPa corrected for the fraction of the electron beam energy deposited in the gas using the data shown in Fig. 7.

calculate reflection coefficients for normal incidence. The following values were obtained for the maxima of the second continua: 0.27 for Ne and Ar, 0.26 for Kr, and 0.24 for Xe.

The analysis of reflection on the foil and its frame is complicated by the fact that the foil bulges due to the pressure differential between the gas cell and the evacuated electron gun. Since at high pressures a considerable part of the light emission originates from the volume directly in front of the foil,[29] focusing/defocusing effects due to the bulging have to be evaluated. However, as discussed in the next paragraph, these effects appear to be negligible. Therefore, in the calculations of the physical efficiency, the foil is assumed to be flat.

Optical measurements of the bulging have given 20 ± 10 μm for the center of the 0.7×0.7 mm^2 foil at 1000 hPa pressure differential. This value is in good agreement with the corresponding value calculated using published foil deformation data:[32] 10–15 μm for the 0.5 $\times0.5$ mm^2 foil, 15–25 μm for the 0.7×0.7 mm^2 foil, and $\sim$75 μm for the 2×2 mm^2 foil. Using the spatial distribution of the light emission provided by CASINO simulations and adopting formulas for the foil bending shape,[32] detailed ray tracing calculations were performed. The calculations provided the integrated light intensity on the detector due to the reflection on the foil and its flat frame. The calculations show that the foil slightly defocuses light. The effect is strongest for Xe at the highest pressure (1400 hPa), when the light emitting volume has the smallest size: The intensity of the reflected light which reaches the detector is reduced to about 80% from the value obtained for the flat configuration. However, since the reflection coefficient is 0.24, the overall effect of the defocusing is quite small: The sum of intensities of the direct light from the gas and the reflected light is reduced by less than 4% if the defocusing effects are taken into account. Experiments performed with foil sizes of 0.5×0.5, 0.7×0.7, and 2×2 mm^2 have shown that the energy conversion efficiency practically does not depend on the foil size (and, therefore, on the bulging profile) for all gases and pressures, strongly suggesting that the focusing/defocusing effects can indeed be neglected.

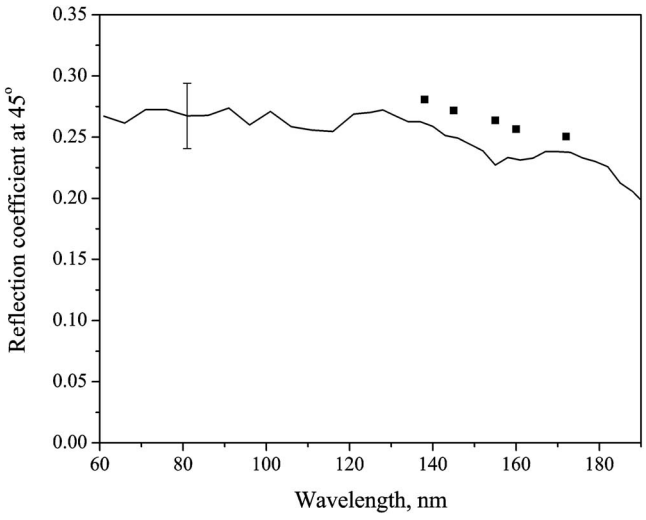


FIG. 9. Reflection coefficient for a flat silicon nitride surface (45° angle of incidence). Solid line: measured data. Dots: calculated values.

The last factor to be considered is the variation in the size of the light emitting volume with gas pressure. As we have shown before,[29] the size of the light emitting volume increases with decreasing gas pressure. Also, the larger the atomic number of the target gas, the smaller is the characteristic size at the same pressure. Therefore, the minimum gas pressures used in this study were 75 hPa for Xe, 100 hPa for Kr, 150 hPa for Ar, and 400 hPa for Ne. These pressures were chosen to ensure that energetic electrons which excite the gas do not reach the side walls of the gas cell and that the apertures used to stop the reflected light were not blocking the direct light to the detector.

The size of the light emitting volume has several effects on the detected light intensity. Due to the enlargement in the light emitting volume, the point-source approach to calculate the total light emission intensity using Eq. (1) becomes inaccurate. However, simulations which use the spatial distributions supplied by CASINO show that this effect is weak. Application of the corresponding corrections decrease the efficiencies calculated for the point source by less than 3% of the uncorrected value for all experimental conditions used in this study.

Also, at reduced pressures, a considerable part of the light emitting volume can extend beyond the 5×5 mm^2 area in front of the foil and its frame and thus the reflection intensity would be reduced. This effect is relatively strong at low pressures for all the gases (results in up to 10% correction in the efficiency values) and has to be applied for gas pressures typically up to twice the minimum pressures.

By applying all the corrections discussed above to the apparent conversion efficiencies, we obtain the values of physical efficiency presented in Fig. 10. The error bars are the systematic uncertainties which are calculated by summing in quadrature the uncertainties due to the following factors: Emission intensity measurements (10% for Ne, 5% for Ar, and 3% for Kr and Xe), beam current measurements (3%), calculations involving reflection of the light on the foil

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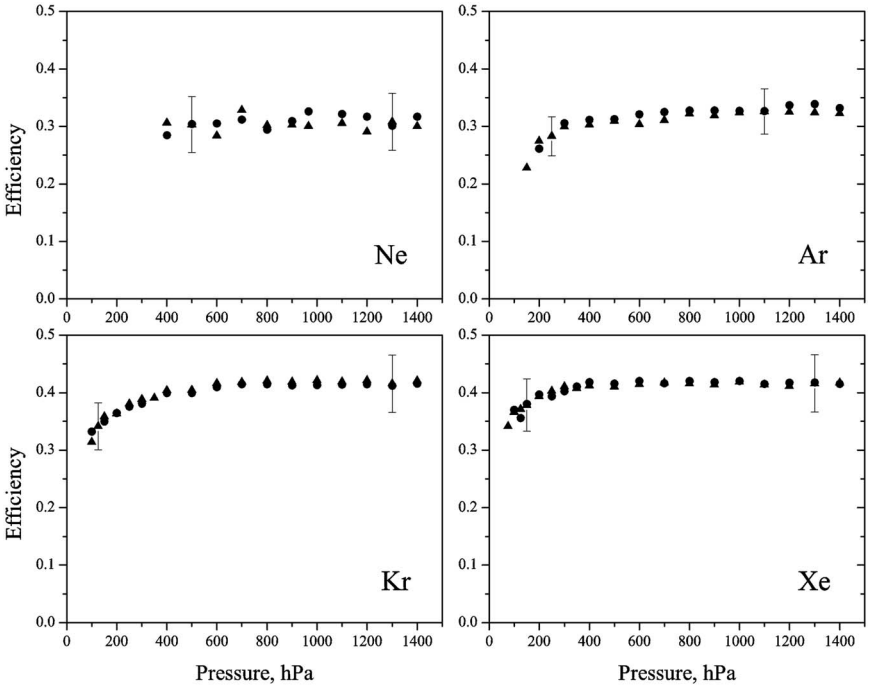


FIG. 10. Conversion efficiencies of the energy deposited by the electron beam in the gas to VUV light for Ne, Ar, Kr, and Xe as a function of gas pressure.

(5%), GEANT4 simulations of the energy deposition in the gas (10%), and a 5% uncertainty for Ne due to the hydrogen impurity. The uncertainty in the GEANT4 simulations has been estimated based on the results presented in Fig. 8: The corrected efficiency versus particle energy of the electron beam calculated using GEANT4 simulations varies by less than 10% (see the discussion above). A more accurate estimation of this uncertainty would require dedicated experiments.

The results of this study show that the energy conversion efficiencies for continuous electron beam excitation are practically constant in the pressure range from 500 to 1400 hPa and have values of 0.31 ± 0.06 for Ne, 0.33 ± 0.04 for Ar, 0.42 ± 0.05 for Kr, and 0.42 ± 0.05 for Xe. The efficiencies show no variation with the electron beam power in the investigated range of 5–50 mW for a fixed particle energy of the electron beam. The presented efficiency values are close to the theoretical maxima of 0.46 for Kr and 0.43 for Xe calculated for electron beam excitation.[13,14] Note that in an earlier publication,[17] theoretical maxima of 0.53 for Ar, 0.52 for Kr, and 0.54 for Xe have also been obtained. Previous experimental measurements of the efficiencies which can be found in literature show values of ~0.46 for Kr (Ref. 14) and ~0.50 for Xe,[12,13] reported for high power pulsed electron beam excitation. To our best knowledge, no efficiency data have been published on the efficiency of the XUV emission from Ne. The efficiency data presented in this paper can contribute to both light source development and further theoretical understanding of the interaction of continuous low energy electron beams with rare gases.

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